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Invited review article

Extrathyroidal extension—what does it mean

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ABSTRACT

Prognostic factors and risk group analysis are very well known in well differentiated thyroid cancer. One of the major prognostic factors is presence of extrathyroidal extension and residual gross tumor. It is not uncommon to find the final pathology report showing minimal extrathyroidal extension. The debate continues as to the implication of this microscopic or minimal extrathyroidal extension. The previous staging system had upstaged such tumors. However, several reports in the literature suggest no prognostic implication of microscopic or minimal extrathyroidal extension.

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Park et al. conducted an interesting review of their data, analyzing patients undergoing total thyroidectomy for extrathyroidal extension [1]. They appropriately divided patients into three groups: no extrathyroidal extension, microscopic extrathyroidal extension, and macroscopic extrathyroidal extension. The authors found differences among the groups, with patients with macroscopic extrathyroidal extension experiencing the worst outcomes. This information has been well-known to thyroid surgeons for some time. However, the authors have done a commendable job of reviewing their data on the different types of extrathyroidal extension. Of interest, they concluded that minimal extrathyroidal extension has some influence on overall outcomes and recurrence-free survival. While these findings are interesting, there are a few points that strike the reader when reviewing this manuscript.

1. A large number of patients in this series had microcarcinomas. There has been a trend in the United States to choose continuous observation over initial surgery for patients with microcarcinomas. This has been well-described in the Japanese literature by Ito and Miyauchi [2]. Our experience at Memorial Sloan Kettering (MSK) mirrors the experience from Japan.
2. The authors do not mention histologic variations among these tumors. There are clearly differences between well-differentiated, tall cell, insular, and poorly differentiated thyroid carcinomas in terms of recurrence.
3. The authors give data on recurrences in terms of type of extrathyroidal extension; however, they do not mention the location of the recurrences. Extrathyroidal extension is implicated more commonly in local recurrences than in nodal recurrences. Multiple factors influence nodal recurrences in well-differentiated thyroid carcinomas, including size of the tumor, initial and follow-up evaluation, how the recurrence was identified, biochemical or structural characteristics, and commonality of nodal metastasis. Although the presence of extrathyroidal extension is a concern, it does not appear to be directly related to nodal recurrence. It does, however, clearly have a major influence on local recurrence and overall outcomes.
4. For gross or macroscopic extrathyroidal extension, the critical aspect with regard to recurrence and outcomes is whether all of the tumor was removed. The most important prognostic factor in the surgical management of patients with thyroid cancer is whether gross tumor was left behind. Patients in whom all gross tumor was not removed have a very high incidence of local recurrence and worse overall outcomes.

Prognostic factors and risk group stratification for thyroid cancer have been well-described. Prognostic factors were initially described by Byar, from France, in 1979 [3]. Subsequently, several publications from around the world have identified more or less the same prognostic factors, which include age, grade of the tumor, extrathyroidal extension, distant metastasis, and invasiveness [4]. Of these, the presence of extrathyroidal extension is perhaps the most important. However, detailed analysis of the degree of extrathyroidal extension has not been performed by many institutions.

Hay et al., from the Mayo Clinic, initially termed their prognostic factors AGES, (age, grade of the tumor, extrathyroidal extension, and size of the tumor) [5]. A subsequent publication from these authors discussed prognostic factors as MACIS (distant metastasis, age, completeness of resection, invasion, and size of the tumor), with completeness of resection a major prognostic factor [5]. For local recurrence among patients with well-differentiated thyroid cancer, completeness of resection is probably the most important

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prognostic factor. The most significant concern for these patients is local recurrence, rather than nodal metastasis. At MSK, we categorized the prognostic factors as GAMES (grade of the tumor, age, distant metastasis, extrathyroidal extension, and size of the tumor) [6]. After critical review of these prognostic factors, we devised a system to stratify patients into low-, intermediate-, and high-risk groups [ref]. This risk group stratification has had a significant influence on overall outcomes among patients with thyroid cancer, as well as on the overall philosophy governing adjuvant treatment in these patients. The risk group stratification that we described is now used by the American Thyroid Association (ATA) to differentiate patients into low-, intermediate-, and high-risk groups. The ATA guidelines appropriately include pathologic features in prognosis. Again, the presence of gross extrathyroidal extension and the completeness of resection are of significant importance.

There has been considerable debate over the years regarding the most appropriate treatment for patients with minimal extrathyroidal extension or involvement of the perithyroidal adipose tissue on final pathologic report. The most important factor for such patients is whether there is gross involvement of the extrathyroidal musculature or minimal involvement with extrathyroidal extension of the disease. It is not uncommon to see minimal extrathyroidal extension in tumors > 2 cm in size. As the average thickness of the thyroid gland is 1–1.5 cm, a tumor larger than that will be located under the surface of the strap muscles. In my practice, I prefer to resect the sternothyroid muscle as an oncologic cushion in patients with suspected or proven thyroid cancer. However, several series in the literature have shown that minimal extrathyroidal extension does not portend worse prognosis [7,8]. Nixon et al. and Hay et al., from MSK and the Mayo Clinic, respectively, reviewed large series of patients with minimal extrathyroidal extension and found it has no effect on long-term outcomes [7,8].

Authors in this manuscript discuss stage migration to T3 for patients with minimal extrathyroidal extension [1]. However, it should be noted that this stage migration was described in the 6th and 7th editions of the AJCC staging system. The 8th edition, which was published in January 2017, revised the stage grouping and does not consider minimal extrathyroidal extension to be a reason for T stage migration. This is in response to having minimal extrathyroidal extension without prognostic importance. In the past, adjuvant radioactive iodine treatment was commonly used for patients with minimal extrathyroidal extension. In the authors' series, 92% of patients received radioactive iodine ablation. The majority of thyroid cancer practitioners today do not routinely use radioactive iodine in this group, in accordance with the 2015 version of the ATA guidelines. It has been concluded that radioactive iodine has very little benefit for patients with minimal extrathyroidal extension [9]. Obviously, for patients with gross extrathyroidal extension or with tumor left behind, radioactive iodine is likely appropriate, and for selected patients with aggressive histologic subtypes, external radiation therapy should be considered. However, the routine use of radioactive iodine for patients with minimally invasive extrathyroidal extension is generally not indicated. The new staging system, although published in January 2017, will be officially implemented in January 2018, at which time T staging will change.

The treatment of patients with thyroid cancer has generated considerable debate and controversy, initially in relation to extent of thyroid surgery and now in relation to prognostic factors, the use of radioactive iodine ablation, and prophylactic central compartment dissection. It is important to remember that the majority of patients with well-differentiated thyroid cancer do remarkably well, with long-term survival exceeding 98%. In addition, we should not neglect the importance of certain factors of poor prognosis—in particular, aggressive histologic subtype, large tumor size, gross extrathyroidal extension, and gross tumor left behind. These factors play a critical role in decision-making and have considerable influence on outcomes.

Concerns over extrathyroidal extension should be raised when considering the initial surgical procedure. The operating surgeon should evaluate for extrathyroidal extension, both preoperatively with appropriate imaging and intraoperatively based on gross findings. If gross extrathyroidal extension is identified (on the basis of clinical evaluation or recurrent laryngeal nerve palsy), the patient should be appropriately imaged with ultrasound and CT scan with contrast. For some time, the belief persisted that CT with contrast should not be used for these patients, on the basis that it would delay radioactive iodine ablation with iodinated contrast. However, there is now general agreement among surgeons that CT scan with contrast is crucial to determine the extent of extrathyroidal extension and the location of the extrathyroidal extension and to ensure appropriate decision-making regarding surgical choice.

Appropriate surgical treatment depends on the specific presentation of the extension. Anterior extrathyroidal extension involving the strap muscles, for example, can be easily surgically resected with negative margins (R0). The new staging system does distinguish between anterior and posterior extrathyroidal extension. Tumors involving the recurrent laryngeal nerve and encircling the nerve require resection of the recurrent laryngeal nerve on the ipsilateral side. Before resecting the functioning recurrent laryngeal nerve, it is important to evaluate whether the other nerve can be well-preserved at the conclusion of the procedure. Patients with involvement of the tracheal wall or esophageal musculature represent truly complex surgical cases in terms of both evaluation and choosing the appropriate intervention. Patients with tumors involving the trachea or with gross submucosal or intratracheal extension require appropriate tracheal resection. If the tumor is mildly adherent to the tracheal wall, it can be easily shaved off with excellent outcomes and R0 resection.

Decision-making for patients with extrathyroidal extension involves appropriate preoperative imaging with ultrasound, cross-sectional imaging with CT or MRI, and critical intraoperative evaluation to ensure resection of all gross tumor. As overall outcomes in patients with R0 resection are far superior to those with R1 or R2 resection, the goal of the operating surgeon should always be to achieve R0 resection in patients with thyroid cancer. Although it is an important clinical finding, minimal extrathyroidal extension, as seen on the final pathologic report, does not appear to have major prognostic implication.

Conflict of interest

I have nothing to disclose.
I have no conflict of interest.

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